Multiplicative relationships between terms

Task A: The structure of multiplicative relationships

1) Write the family of rearrangements for the below equations.

a)

$12 \times 7 = 84$	$\frac{84}{7} = 12$
$7 \times 12 = 84$	$\frac{84}{12} = 7$

b)

$0.3 \times 2.7 = 0.81$	$\frac{0.81}{2.7} = 0.3$
$2.7 \times 0.3 = 0.81$	$\frac{0.81}{0.3} = 2.7$

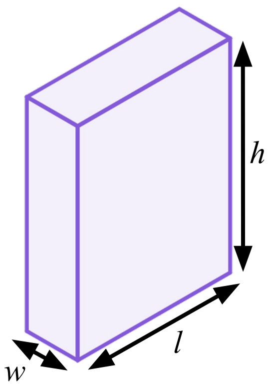
c)

$7 \times y = 3$	$\frac{3}{y} = 7$
$y \times 7 = 3$	$\frac{3}{7} = y$

d)

$d = s \times t$	$\frac{d}{t} = s$
$d = t \times s$	$\frac{d}{s} = t$

2) The volume of a cuboid can be calculated as 'length × width × height'



Fill in the missing parts of these rearrangements.

$V = lwh$

$\frac{V}{h} = lw$ $\qquad$ $\frac{V}{w} = lh$

$\frac{V}{lw} = h$ $\qquad$ $\frac{V}{wh} = l$

*V, the **product** of the original multiplicative relationship remained the numerator.*

Task B: Rearranging multiplicative relationships

1) Write the family of rearrangements for the below equations.

a)

$19 \times 5 = 95$ $\qquad \dfrac{95}{5} = 19$

$5 \times 19 = 95$ $\qquad \dfrac{95}{19} = 5$

b)

$0.2 \times 4.1 = 0.82$ $\qquad \dfrac{0.82}{4.1} = 0.2$

$4.1 \times 0.2 = 0.82$ $\qquad \dfrac{0.82}{0.2} = 4.1$

c)

$7 \times y = 56$ $\qquad \dfrac{56}{y} = 7$

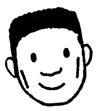
$y \times 7 = 56$ $\qquad \dfrac{56}{7} = y$

d)

$P = F \times A$ $\qquad \dfrac{P}{A} = F$

$P = A \times F$ $\qquad \dfrac{P}{F} = A$

2) Lucas is rearranging equations.



I just 'swap positions' in the equation.

In each case decide if Lucas is correct and write a sentence to justify your decision.

a) $\dfrac{72}{9} = 8$ can swap to $\dfrac{72}{8} = 9$ ✓

Both are rearrangements of $8 \times 9 = 72$ *where* 72 *is the* ***product***.

b) $\dfrac{5}{h} = b$ can swap to $\dfrac{5}{b} = h$ ✓

Both are rearrangements of $bh = 5$ *where* 5 *is the* ***product***.

c) $\dfrac{13}{x} = 2$ can swap to $\dfrac{2}{x} = 13$ ✗

One is a rearrangement of $2x = 13$ *and the other is a rearrangement of* $13x = 2$

3) Rearrange each equation into the form $a \times b = c$

a) $\dfrac{x}{8} = y$

$x \times \dfrac{1}{8} = y$

$x \times \dfrac{1}{8} \times 8 = y \times 8$

$x = 8y$

b) $\dfrac{x}{3y} = y$

$x \times \dfrac{1}{3y} = y$

$x \times \dfrac{1}{3y} \times 3y = y \times 3y$

$x = 3y^2$

c) $\dfrac{5x}{3yz} = 4z^2$

$\dfrac{5x}{3yz} \times 3yz = 4z^2 \times 3yz$

$5x = 12yz^3$